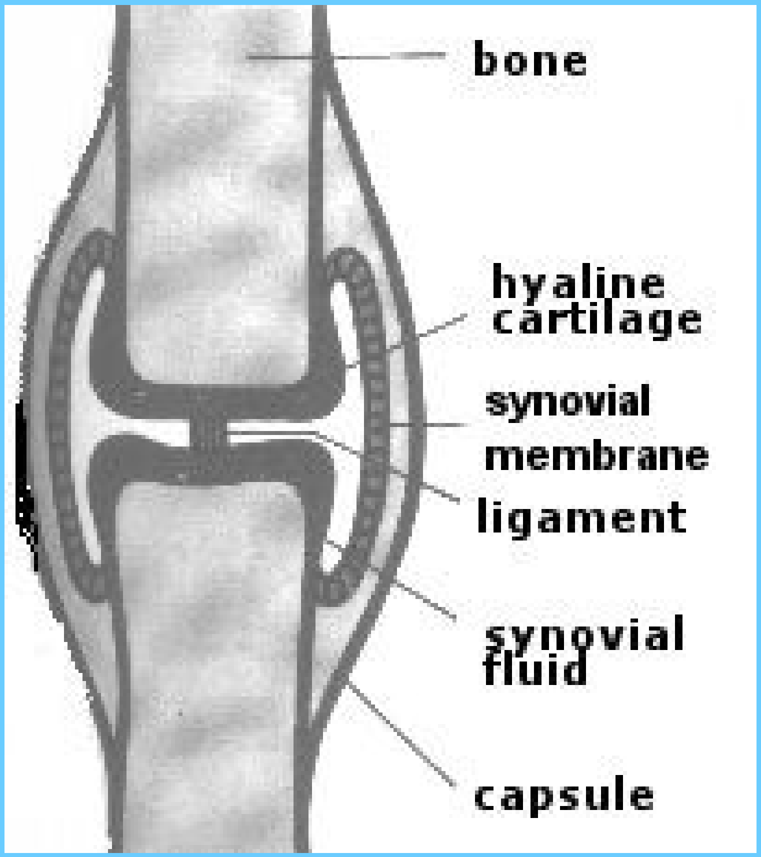
“JOINTS,CLASSIFICATION OF JOINTS,CLASSIFICATION OF SYNOVIAL JOINTS,FEATURES OF SYNOVIAL JOINTS”

LEARNING OBJECTIVE

- *At the end of lecture student must be able to know,*
- *What is joint?,*
- *Classification of joints,*
- *Synovial joint structure,*
- *Nerve supply,*
- *Blood supply,*
- *Movements possible in synovial joint,*
- *Synovial joints properties.*

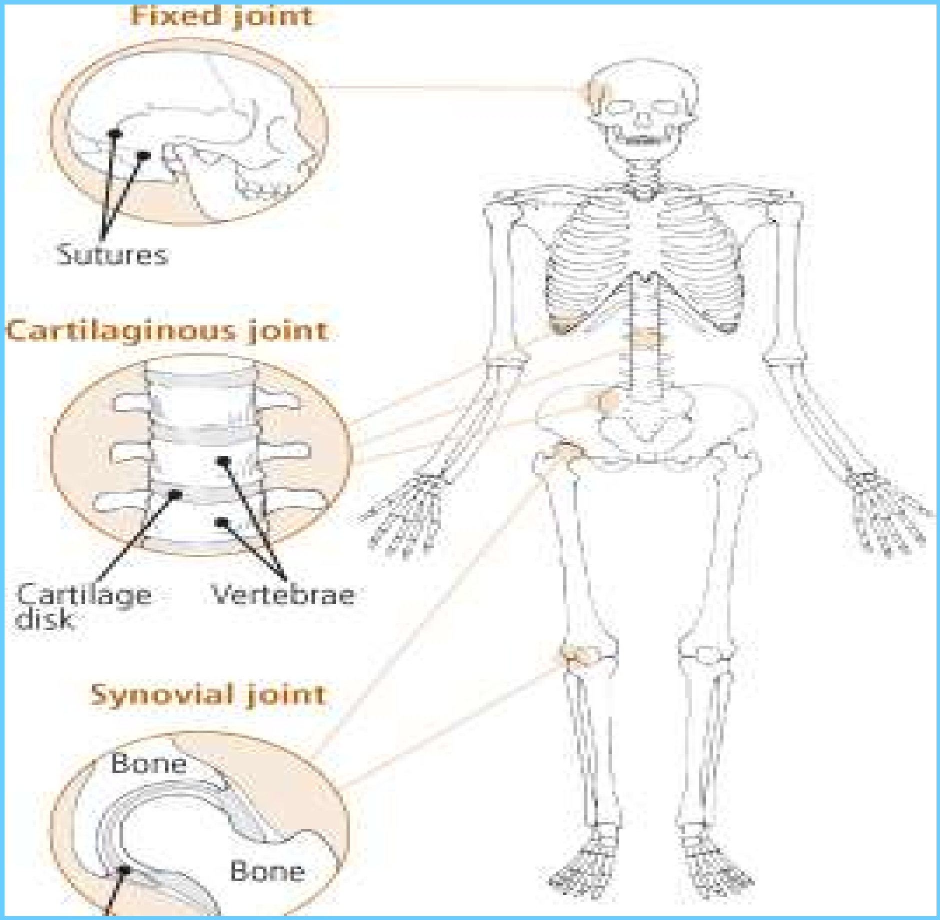
<u>JOINT</u>

- ***A joint is the location at which two or more <u>bones</u> make contact.***
- ***They are constructed to allow movement and provide mechanical support, and are classified structurally and functionally***



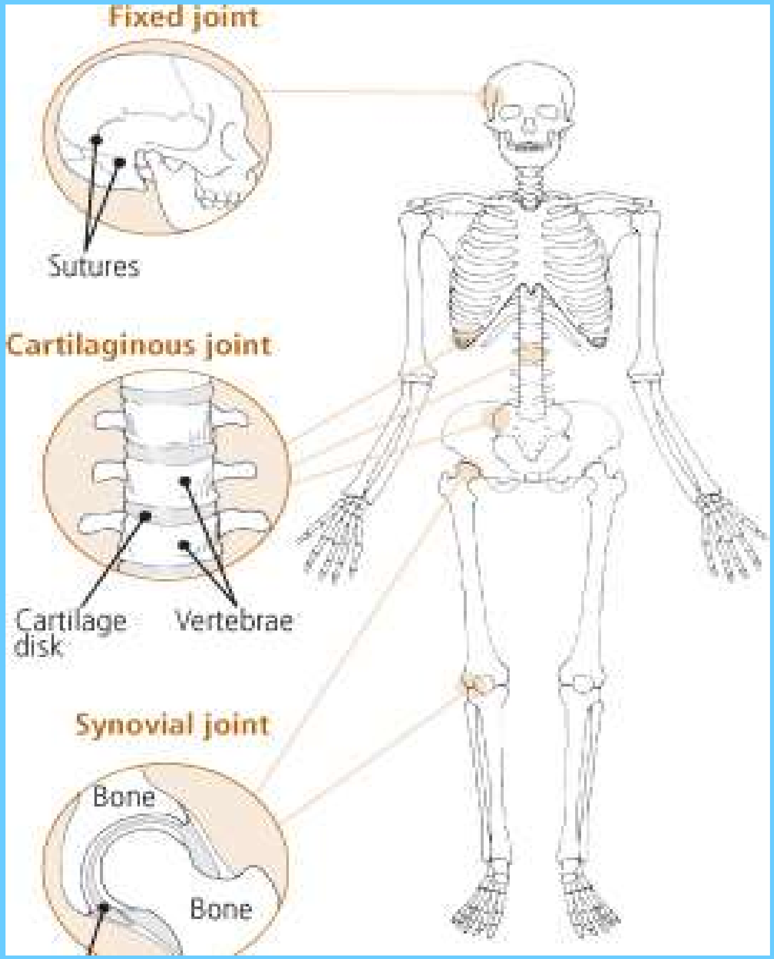
CLASSIFICATION OF JOINTS

- ***Joints are classified structurally and functionally.***
- ***Structural classification is determined by how the bones connect to each other.***
- ***Functional classification is determined by the degree of movement between the articulating bones.***



STRUCTURAL CLASSIFICATION

- *There are three structural classifications of joints:*
- *Fibrous joint - joined by fibrous connective tissue*
- *Cartilaginous joint - joined by cartilage*
- *Synovial joint - not directly joined*



FUNCTIONAL CLASSIFICATION

Joints can also be classified functionally

- ***synarthrosis - permits little or no mobility. Most synarthrosis joints are fibrous joints (e.g., skull sutures).***
- ***amphiarthrosis - permits slight mobility. Most amphiarthrosis joints are cartilaginous joints(e.g., vertebrae).***
- ***diarthrosis - permits a variety of movements. All diarthrosis joints are synovial joints (e.g., shoulder, hip, elbow, knee, etc.), and the terms "diarthrosis" and "synovial joint" are considered equivalent by Terminologia Anatomica***

BIOMECHANICAL CLASSIFICATION

- ***Joints can also be classified based on their anatomy or on their biomechanical properties.***
- ***According to the anatomic classification, joints are subdivided into simple and compound, depending on the number of bones involved, and into complex and combination joints***
- ***Simple Joint: 2 articulation surfaces (eg. shoulder joint, hip joint)***
- ***Compound Joint: 3 or more articulation surfaces (eg. radiocarpal joint)***
- ***Complex Joint: 2 or more articulation surfaces and an articular disc or meniscus (eg. knee joint)***

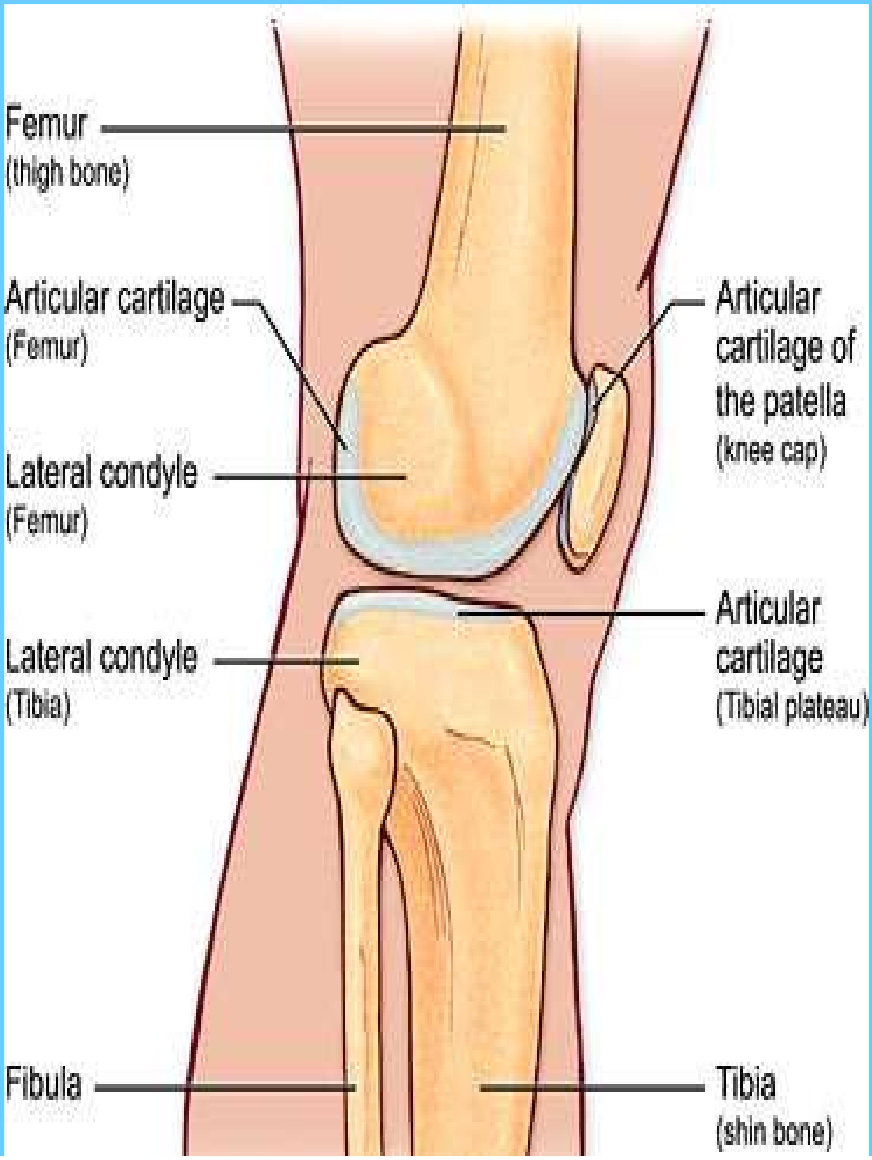
ANATOMICAL

The joints may be classified anatomically into the following groups:

- *Articulations of hand*
- *Elbow joints*
- *Wrist joints*
- *Axillary articulations*
- *Sternoclavicular joints*
- *Vertebral articulations*
- *Temporomandibular joints*
- *Sacroiliac joints*
- *Hip joints*
- *Knee joints*
- *Articulations of foot*

SYNOVIAL JOINT

- ***Is the most common and most movable type of joint in the body of a mammal.***
- ***As with most other joints, synovial joints achieve movement at the point of contact of the articulating bones.***

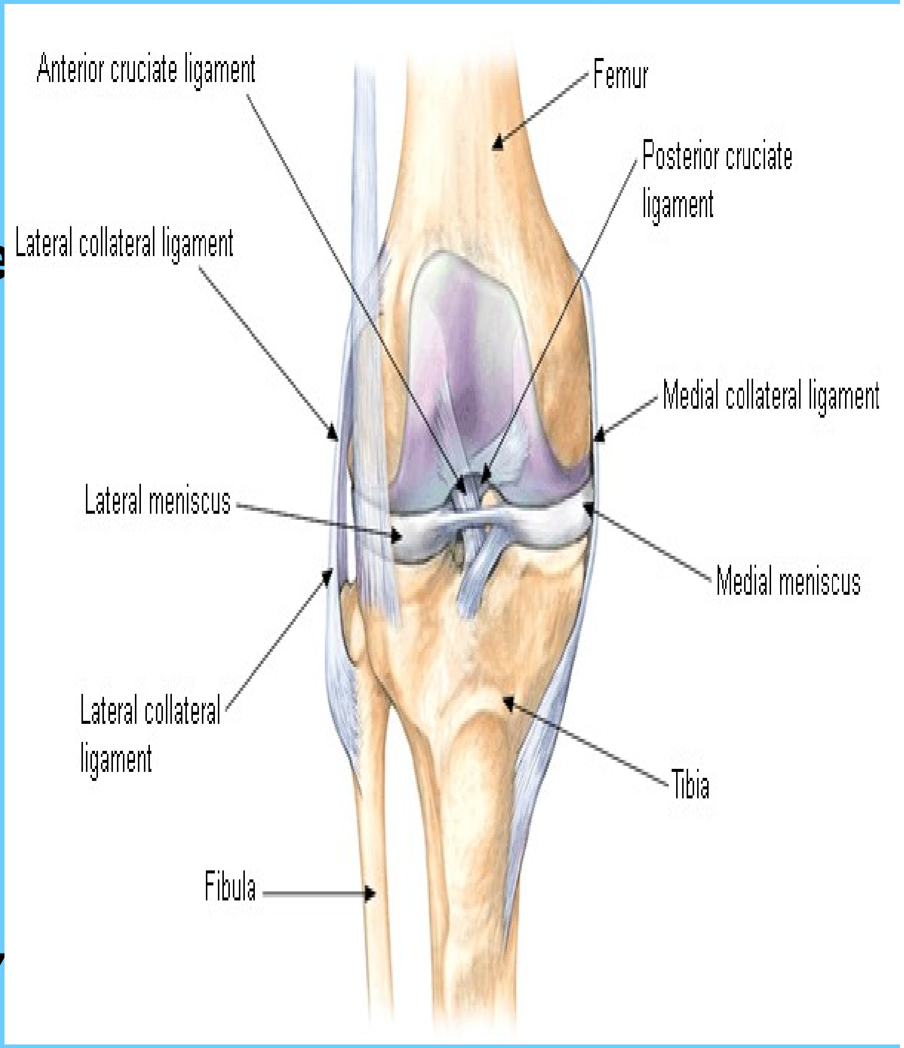


STRUCTURAL AND FUNCTIONAL DIFFERENCES

- ***Distinguish synovial joints fromcartilaginous joints (synchondroses and symphyses) and fibrous joints (sutures, gomphoses, and syndesmoses).***
- ***The main structural differences between synovial and fibrous joints is the existence of capsules surrounding the articulating surfaces of a synovial joint and the presence of lubricating synovial fluid within that capsule (synovial cavity).***

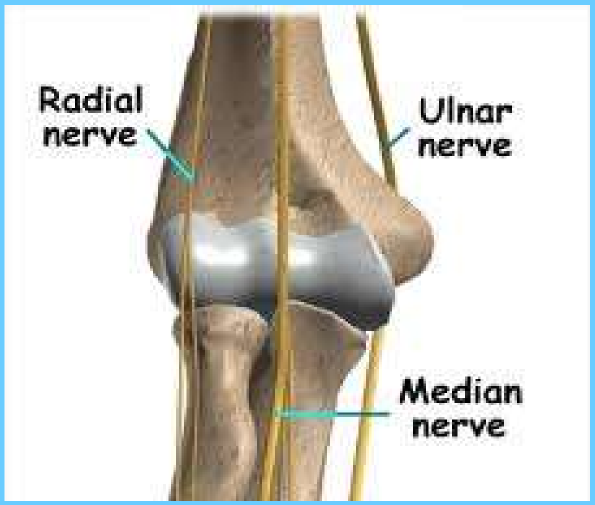
STRUCTURE

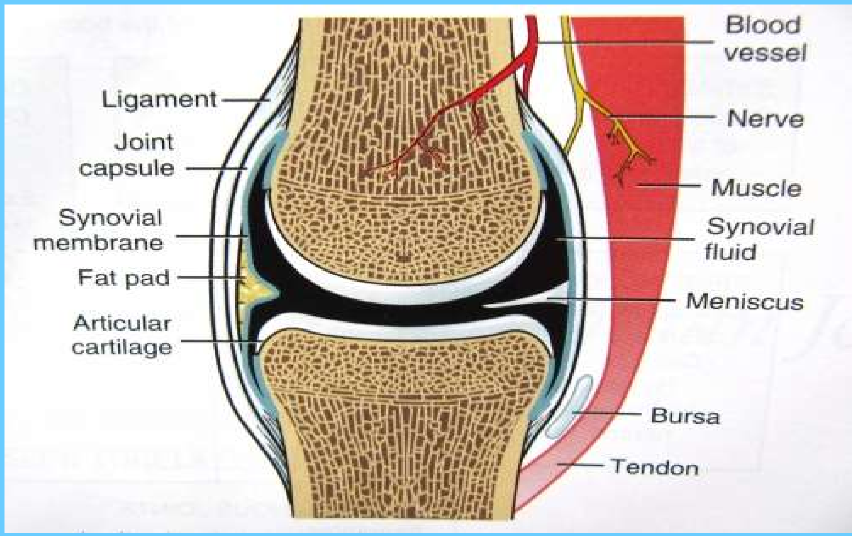
- ***Articular capsule: The fibrous capsule is continuous with the periosteum of bone. It is also highly innervated but avascular (lacking blood and lymph vessels)***
- ***Articular cartilage: lines the epiphyses of joint end of bone. Provides the loading and unloading mechanism to resist load and shock***
- ***Synovial membrane: the inner layer of the fibrous articular capsule. The synovial membrane covers the lining of the synovial cavity where articular cartilage is absent.***



NERVE SUPPLY OF SYNOVIAL JOINT

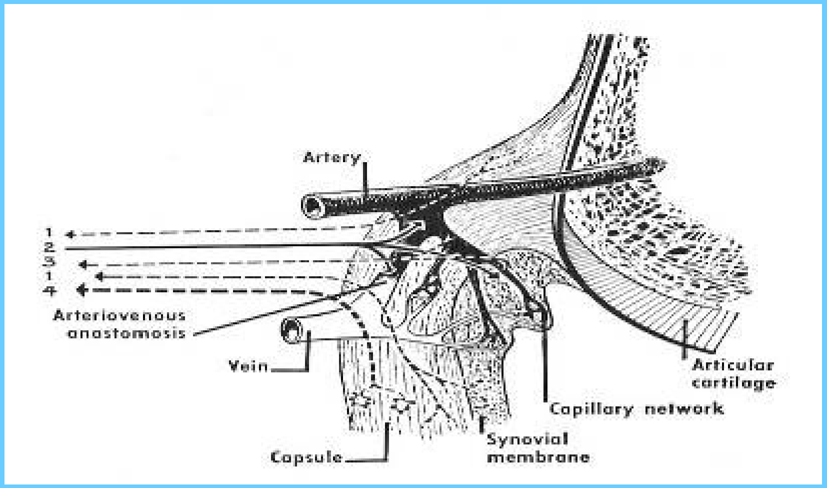
- *It is derived from the nerve supply of muscles acting on the joint.*

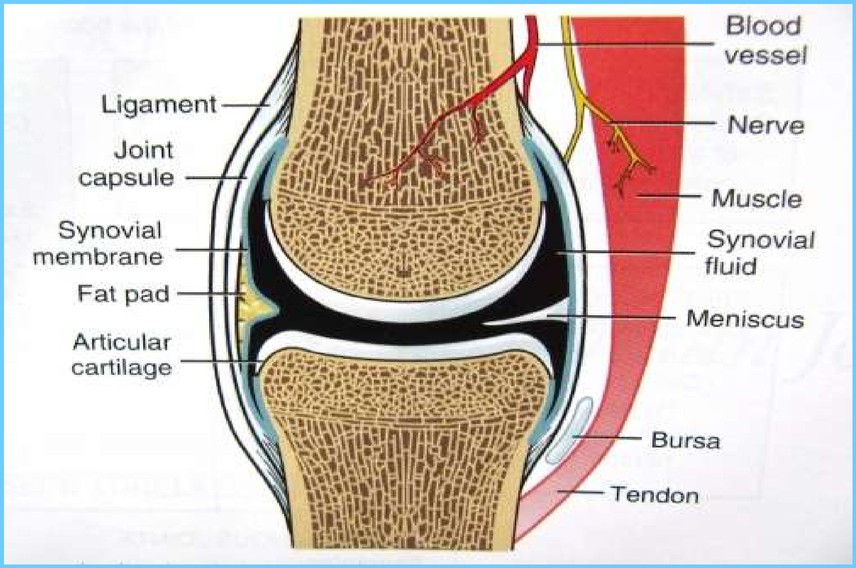




BLOOD SUPPLY OF SYNOVIAL JOINT

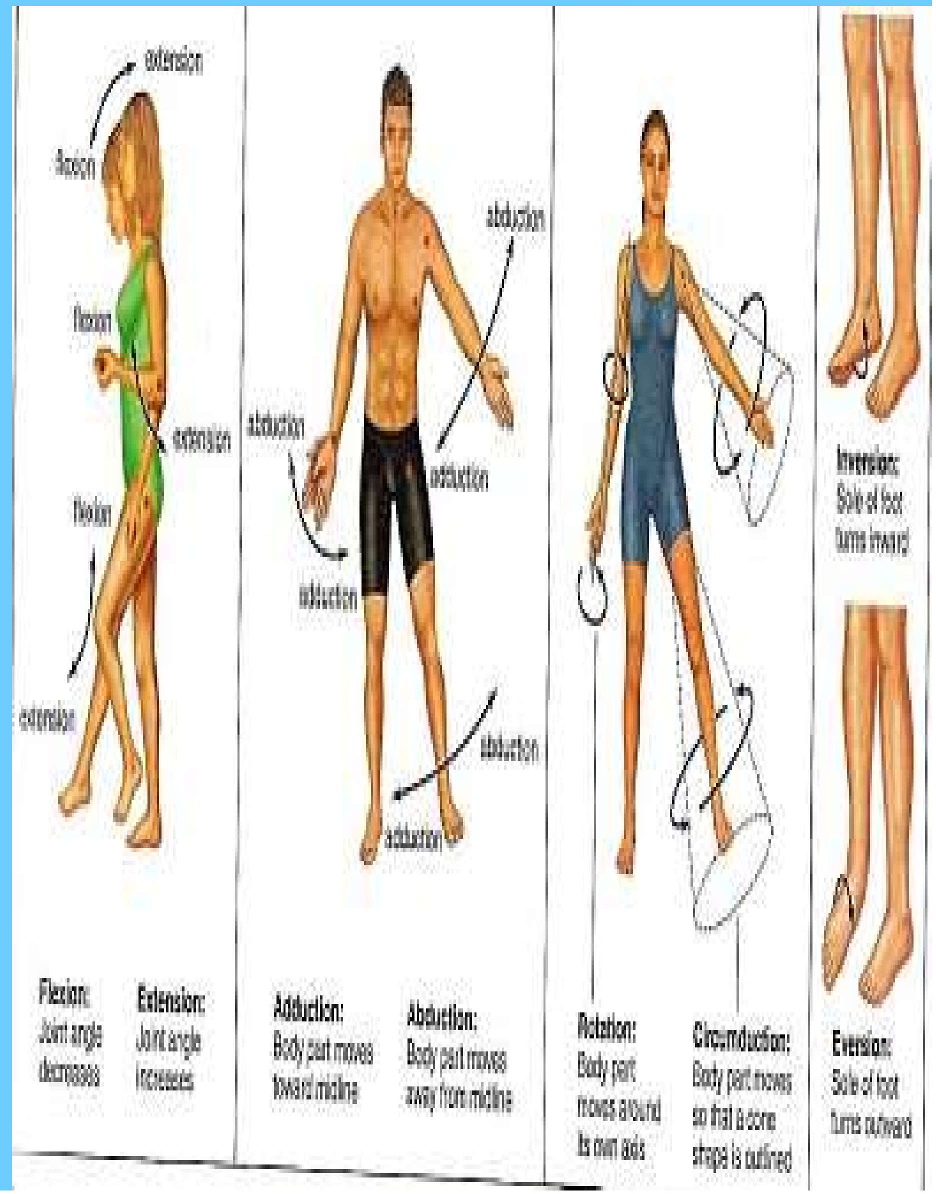
- *From the arteries sharing in the anastomosis aro und the joint.*





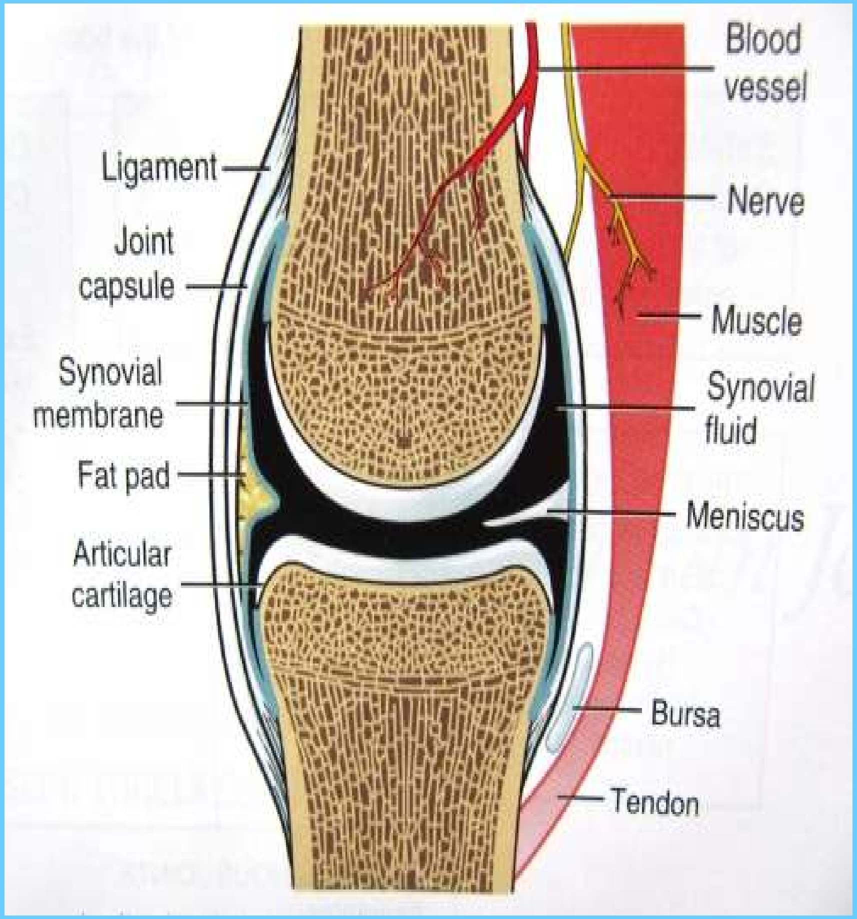
MOVEMENTS POSSIBLE

- ***The movements possible with synovial joints are:***
- ***Abduction: movement away from the mid-line of the body.***
- ***Adduction: movement towards the mid-line of the body.***
- ***Extension: straightening limbs at a joint.***
- ***Flexion: bending the limbs at a joint.***
- ***Rotation: a circular movement around a fixed point.***



FACTORS INFLUENCING JOINT STABILITY

- *The shape of articular surfaces.*
- *capsule & ligaments.*
- *Muscle Tone.*
- *gravity*
- *Atmospheric Pressure.*



SYNOVIAL JOINTS

- There are seven types of synovial joints.
- Some are relatively immobile, but are more stable.
- Others have multiple degrees of freedom, but at the expense of greater risk of injury.
- In ascending order of mobility, they are,
- Gliding joint
- Hingue joint
- Pivoted joint
- Condyloid joints
- Saddle joints
- Ball and socket joints
- Compound joints

GLIDING JOINT(OR PLANAR JOINTS)THE

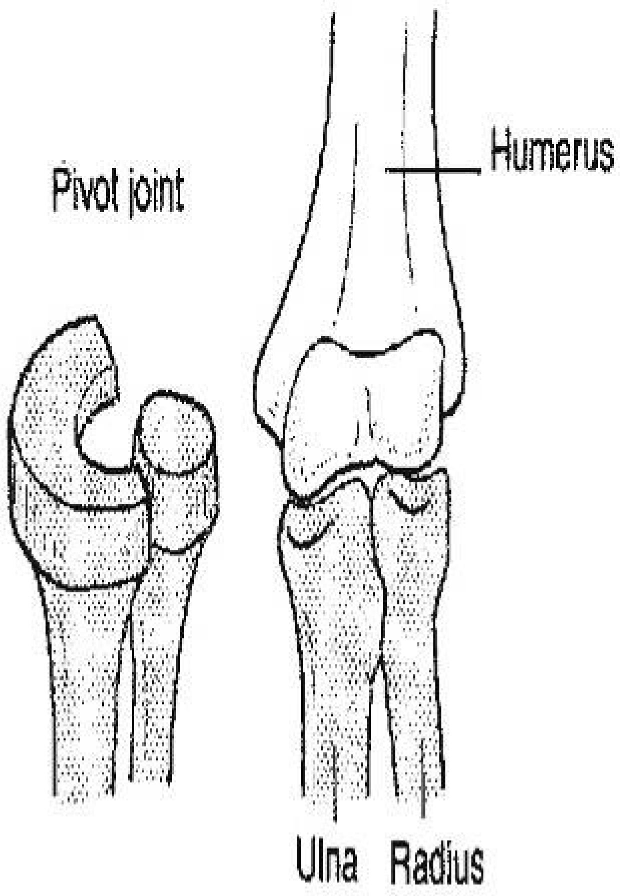
- ***Carpals of the wrist, acromioclavicular joint.***
- ***These joints allow only gliding or sliding movements.***

HINGUE JOINT

- *The elbow (between the humerus and the ulna).*
- *These joints act like a door hinge, allowing flexion and extension in just one plane.*

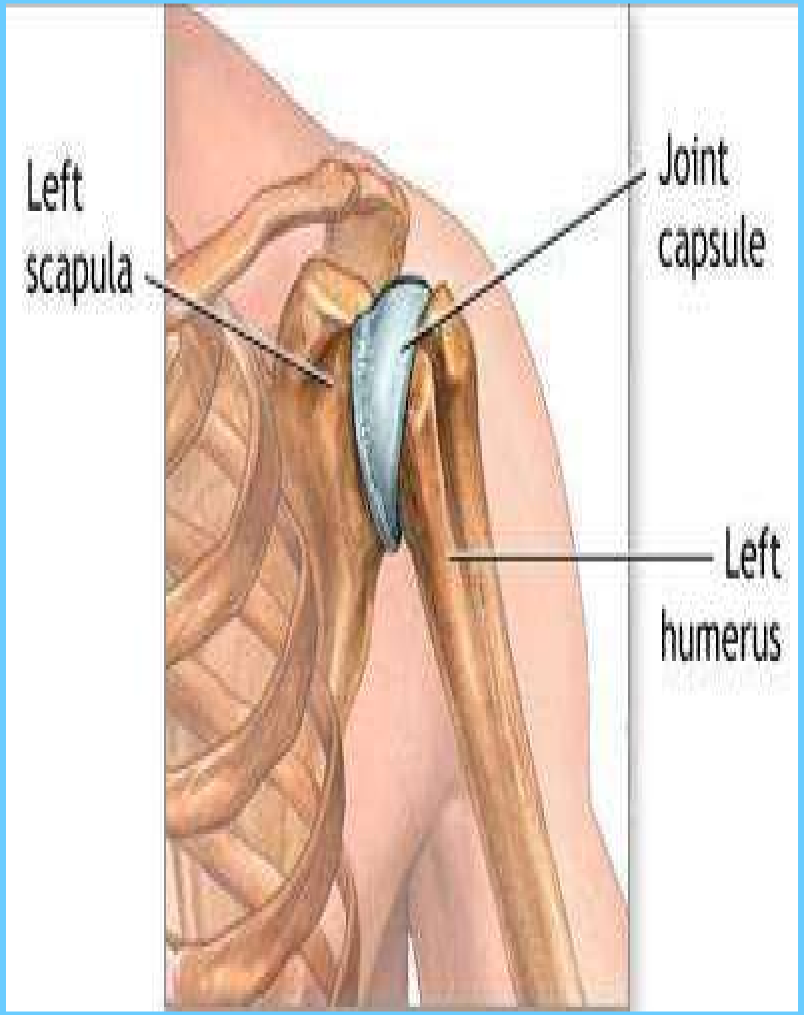
PIVOT JOINTS

- ***Atlanto-axial joint, proximal radioulnr joint, and distal radioulnar joint.***
- ***One bone rotates about another***



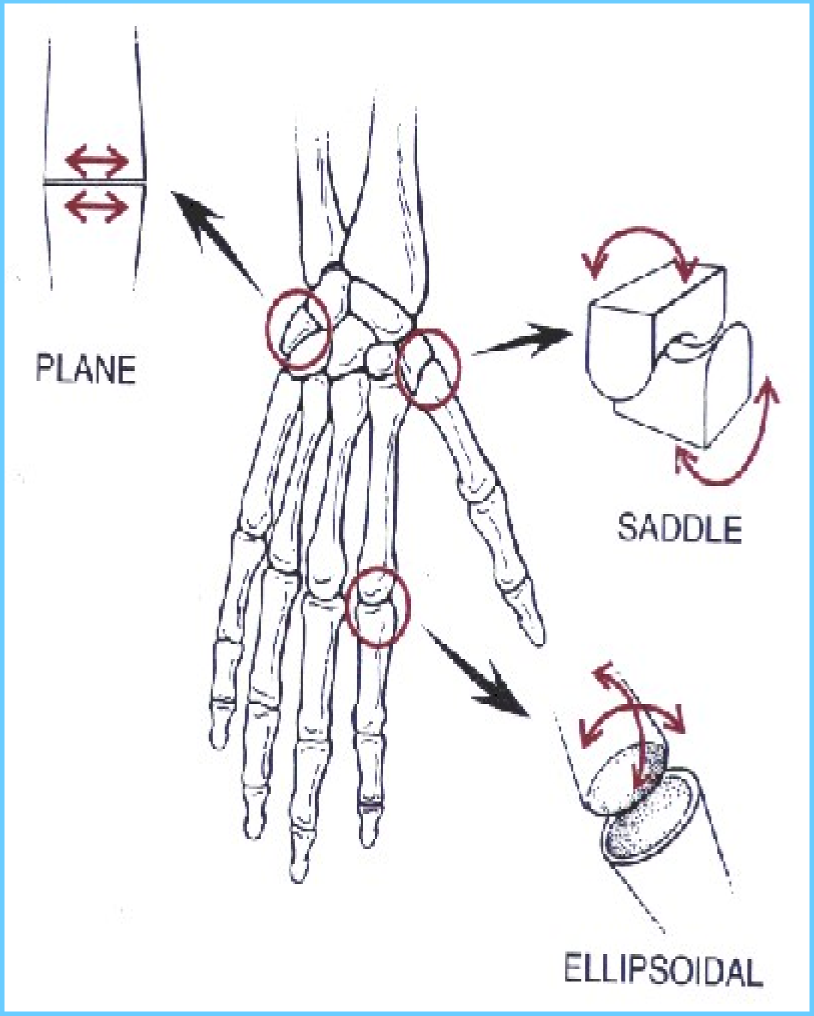
CONDYLOID JOINT

- *(or ellipsoidal joints)the wrist joint (radiocarpal joint).*
- *A condyloid joint is where two bones fit together with an odd shape (e.g. anellipse).*
- *one bone is concave, the other convex.*
- *Some classifications make a distinction between condyloid and ellipsoid joints*



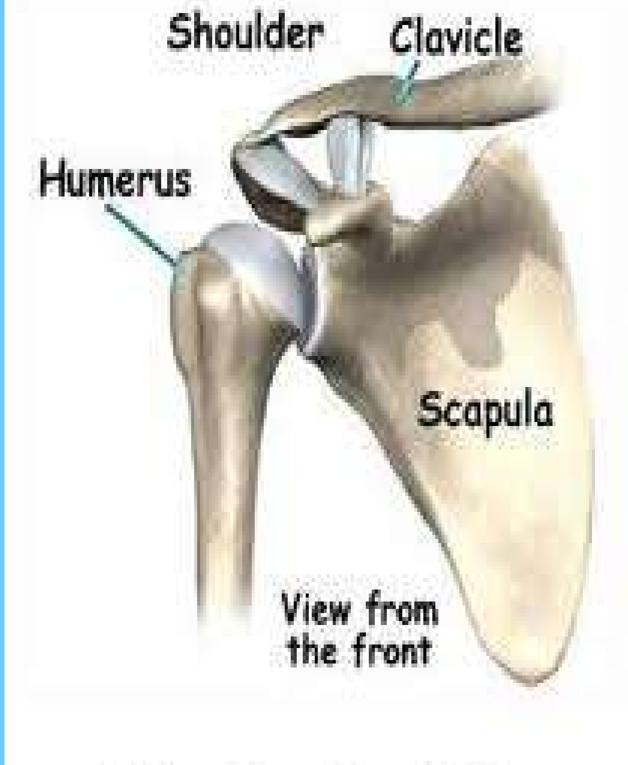
SADDLE JOINT

- ***Carpometacarpal or Trapeziometacarpal Joint of thumb (between the metacarpal and carpal - the trapezium) , sternoclavicular joint.***
- ***Saddle joints, which resemble a saddle, permit the same movements as the condyloid joints.***



BALL AND SOCKET JOINTS

- ***The shoulder(gleno humeral), and hip joints.***
- ***These allow a wide range of movemen***



The shoulder joint

COMPOUND JOINTS

- ***knee jointcondylar joint(condyles of femur join with condyles of tibia) and saddle joint(lower end of femur joins with patela)***

SYNOVIAL JOINTS

- ***Most of your joints are 'synovial joints'. They are movable joints containing a lubricating liquid called synovial fluid. Synovial joints are predominant in your limbs where mobility is important. Ligaments help provide their stability and muscles contract to produce movement. The most common synovial joints are listed below:***
- ***Ball and socket joints, like your hip and shoulder joints, are the most mobile type of joint in the human body. They allow you to swing your arms and legs in many different directions.***
- ***Ellipsoidal joints, such as the joint at the base of your index finger, allow bending and extending, rocking from side to side, but rotation is limited.***

SYNOVIAL JOINTS

- ***Gliding joints occur between the surfaces of two flat bones that are held together by ligaments. Some of the bones in your wrists and ankles move by gliding against each other.***
- ***Hinge joints, like in your knee and elbow, enable movement similar to the opening and closing of a hinged door.***
- ***The pivot joint in your neck allows you to turn your head from side to side.***
- ***The only saddle joints in your body are in your thumbs. The bones in a saddle joint can rock back and forth and from side to side, but they have limited rotation.***

REFERENCES

- ***BD CHAURASIA(CHAURASIA GENERAL ANATOMY,8^{TH} EDITION)***
- ***INTERNET.***